

12 **(long)** A mass m sits at the origin and is connected to fixed plates at $z = \pm \frac{4}{3}l$ by a pair of stretched springs, each with spring constant k and natural length l , as shown in Fig. 12.

(a) How many normal modes do we expect for the mass? Sketch the expected displacement pattern for each mode. [5]

(b) Find the angular frequency, ω_1 , of the normal mode in which the mass moves up and down in the z direction. [5]

(c) If the mass moves in the horizontal ($z = 0$) plane, the change in length of the springs is negligible for small displacements. By using a small angle approximation ($\sin \theta \approx \tan \theta \approx \theta$) or otherwise, show the angular frequency of the normal mode in which the mass moves side-to-side in the x direction is $\omega_2 = \omega_1 / 2$. [10]

(d) The mass is subject to a driving vertical force $\mathbf{f} = f_0 \cos\left(\frac{3}{4}\omega_1 t\right)\hat{\mathbf{z}}$. Find the amplitude and phase of the steady state displacement of the mass relative to the driving force, assuming that damping is negligible once the steady state is achieved. [5]

(e) A viscous fluid is added between the plates that exerts a damping force $\mathbf{f} = -\lambda \mathbf{v}$, where $\mathbf{v}$ is the mass's velocity and λ is the damping constant. If the mass is subject to the same driving force, use the databook to estimate what value of λ would reduce the displacement of the mass by half of its undamped value. [5]

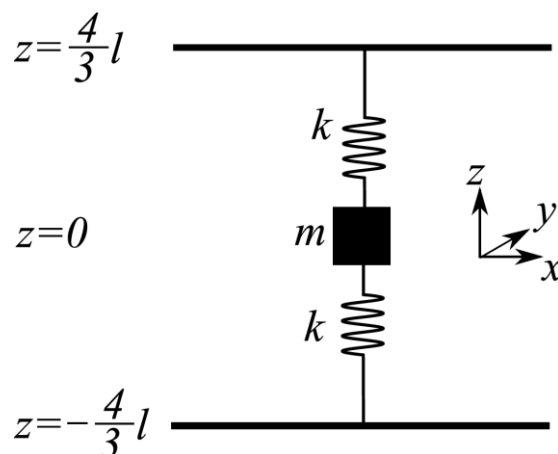


Fig. 12

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